

Ataxic Disorders

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CLINICAL PRESENTATION

- Specific signs:

- ❖ Dysmetria
- ❖ Dyssynergia
- ❖ Intension tremor
- ❖ Dysdiadokokinesia
- ❖ Rebound

- Nonspecific signs:

- ❖ Dysarthria
- ❖ Nystagmus
- ❖ Head nodding
- ❖ Gait ataxia
- ❖ Hypotonia
- ❖ Hyporeflexia or pendular reflex

APPROACH TO THE PATIENT WITH ATAXIA

- Causes are best grouped by determining whether ataxia is symmetric or focal and by the time course
- Also important to distinguish whether ataxia is present in isolation or is part of a multisystem neurologic disorder.
- Acute symmetric ataxia is usually due to medications, toxins, viral infection, or a postinfectious syndrome (especially varicella).
- Subacute or chronic symmetric ataxia can result from hypothyroidism, vitamin deficiencies, infections (Lyme disease, tabes dorsalis, prions), alcohol, other toxins, or an inherited condition.
- An immune-mediated progressive ataxia is associated with antigliadin antibodies; biopsy of the small intestine may reveal villous atrophy of gluten enteropathy.
- Progressive nonfamilial cerebellar ataxia after age 45 suggests a paraneoplastic syndrome, either subacute cortical cerebellar degeneration (ovarian, breast, lung, Hodgkin's) or opsoclonus-myoclonus (neuroblastoma, breast, lung).

- Unilateral ataxia suggests a focal lesion in the ipsilateral cerebellar hemisphere or its connections.
- An important cause of acute unilateral ataxia is stroke.
- Other diseases producing asymmetric or unilateral ataxia include tumors, multiple sclerosis, progressive multifocal leukoencephalopathy, and congenital malformations.

Symmetric and Progressive Signs			Focal and Ipsilateral Cerebellar Signs		
Acute (Hours to Days)	Subacute (Days to Weeks)	Chronic (Months to Years)	Acute (Hours to Days)	Subacute (Days to Weeks)	Chronic (Months to Years)
Intoxication: alcohol, lithium, diphenylhydantoin, barbiturates (positive history and toxicology screen)	Intoxication: mercury, solvents, gasoline, glue; cytotoxic chemotherapeutic drugs	Paraneoplastic syndrome	Vascular: cerebellar infarction, hemorrhage, or subdural hematoma	Neoplastic: cerebellar glioma or metastatic tumor (positive for neoplasm on MRI/CT)	Stable gliosis secondary to vascular lesion or demyelinating plaque (stable lesion on MRI/CT older than several months)
Acute viral cerebellitis (CSF supportive of acute viral infection)	Alcoholic-nutritional (vitamin B ₁ and B ₁₂ deficiency)	Anti-gliadin antibody syndrome	Infectious: cerebellar abscess (mass lesion on MRI/CT, history in support of lesion)	Demyelinating: multiple sclerosis (history, CSF, and MRI are consistent)	Congenital lesion: Chiari or Dandy-Walker malformations (malformation noted on MRI/CT)
Postinfection syndrome	Lyme disease	Hypothyroidism		AIDS-related multifocal leukoencephalopathy (positive HIV test and CD4+ cell count for AIDS)	
		Inherited diseases			
		Tabes dorsalis (tertiary syphilis)			
		Phenytoin toxicity			

EVALUATION

- Diagnostic approach is determined by the nature of the ataxia.
- For symmetric ataxias, drug and toxicology screens; vitamin B1, B12, and E levels; thyroid function tests; antibody tests for syphilis and Lyme infection; anti-gliadin antibodies; paraneoplastic antibodies; and CSF studies often indicated.
- Genetic testing is available for many inherited ataxias.
- For unilateral or asymmetric ataxias, brain MRI or CT scan is the initial test of choice.

Treatment

- The most important goal is to identify treatable entities including hypothyroidism, vitamin deficiency, and infectious causes.
- Parainfectious ataxia can be treated with glucocorticoids.
- Ataxia with anti-gliadin antibodies and gluten enteropathy may improve with a gluten-free diet.
- Paraneoplastic disorders may improve following removal of the tumor or immunotherapy.
- Vitamins B1 and B12 should be administered to pts with deficient levels.
- There is no proven therapy for any of the autosomal dominant ataxias; family and genetic counseling are important.
- Cerebellar hemorrhage and other mass lesions of the posterior fossa may require emergent surgical treatment to prevent fatal brainstem compression.

Inherited ataxia syndromes

Congenital ataxias

- This group of disorders, as the name implies, encompasses ataxic disorders of early onset; generally these are non-progressive.
- There are a number of well-recognized syndromes with a range of additional features including involvement of other organs and dysmorphism.
- Frequently, the history is that of a child, who although able to acquire new skills is always behind their peers and described as clumsy.
- Imaging can sometime help in some syndromes and there may be significant reduction in cerebellar volume, including signs such as ‘molar tooth’ an abnormally deep interpeduncular fossa with elongated, thick and maloriented superior cerebellar peduncles as seen with Joubert’s syndrome.
- The combination of a very small cerebellum and associated structures on imaging in a child who is not profoundly ataxic points strongly towards a congenital explanation.

Congenital ataxias

Syndrome	Additional features	Mode of inheritance	Gene defect
Joubert's syndrome	Episodic hyperpnoea, abnormal eye movements and mental retardation	Autosomal recessive Genetically heterogeneous	<i>AHI1</i> gene <i>NPHP1</i> gene <i>CEP290</i> Plus others with established and distinct loci
Gillespie's syndrome	Mental retardation and partial aniridia	Autosomal recessive	Pax 6
Congenital nystagmus	Hypoplasia of the macula in some cases	Autosomal recessive and X-linked	<i>NYS1</i> –6p <i>NYS2</i> , X-linked and others
Congenital hypoplasia and quadripedal gait	Mental retardation and seizures	Autosomal recessive	17p
Paine's syndrome	Spasticity, mental retardation and microcephaly	X-linked recessive ataxia	No gene identified

Autosomal recessive cerebellar ataxias

- There is a long list of mutations inherited in an autosomal recessive manner and, as a general rule, onset of the autosomal recessive ataxias is before the age of 20 years.
- FRDA is by far and away the most common cause, accounting for approximately 40% of autosomal recessive cerebellar ataxia (ARCA) cases.

Syndrome	Gene defect	Clinical pointers	Notes
Friedreich's ataxia (FRDA)	GAA repeat (and some point mutations in <i>FRDA</i> gene)	Neuropathy, pyramidal signs, skeletal abnormalities, diabetes and cardiomyopathy	Most common ARCA
Ataxia telangiectasia (AT) AT-like disorder	<i>ATM</i> <i>hMRE11</i>	Oculomotor apraxia, mixed movement disorder, humoral immune deficiencies, increased cancer risk	Telangiectasia not always present or easily identified
Cockayne's syndrome	CS type A – <i>ERCC8</i> gene CS type B – <i>ERCC6</i> gene	'Cachectic dwarfism' Mental retardation Pigmentary retinopathy	
Xeroderma pigmentosum (XP)		Increased skin cancer in XP but not Cockayne's	
AOA1	Aprataxin	Oculomotor apraxia	
AOA2	Senataxin	Oculomotor apraxia	Probably second most common ARCA
ARSACS	Sacsin	Demyelinating neuropathy, hypertrophy of the retinal layer (detected using OCT)	Along with senataxin second in frequency to FRDA
Hypogonadism	RNF216	Hypogonadotrophic hypogonadism	a.k.a. Gordon Holmes' syndrome
Marinesco–Sjögren's syndrome	<i>SIL1</i> on chr 5q31	Cataracts and mental retardation	
Progressive myoclonic ataxia (Ramsay-Hunt syndrome)	Genetically complex	Epilepsy is frequently associated	Overlaps with differential of progressive myoclonic epilepsy
Behr's and related syndromes, e.g. 3-methylglutaconic aciduria type III (Costeff's syndrome)	No gene for Behr's yet identified <i>OPA3</i> gene	Optic atrophy (OA), spasticity and mental retardation	There are other cases with OA but not Behr's syndrome
Congenital or childhood onset deafness	Genetically complex	Several different families/ syndromes reported	May cause overlap with Usher's syndrome
Autosomal recessive late onset ataxia	Heterogeneous	Clinically variable with a range of additional features	Relatively rare

Friedreich's ataxia

- This has an estimated prevalence of approximately 2/100 000.
- The condition is characterized by a progressive gait and limb ataxia and a number of additional features including an axonal sensorimotor neuropathy, pyramidal tract involvement, hypertrophic cardiomyopathy, skeletal abnormalities, optic atrophy, deafness and diabetes.
- Typically, it begins between the ages of 8 and 15 years.
- The predominant mutation is a trinucleotide repeat (GAA) in intron 1.
- Expansion of both alleles is found in over 96% of patients.
- The remaining patients have point mutations on one allele in association with an expansion on the other.

Ataxic disorders associated with defective DNA repair

- There are several rare disorders characterized at a molecular level by a reduced capacity to repair DNA. The most well known and most studied is ataxia telangiectasia (AT).
- This produces a mixed movement disorder with the presence of dystonia and chorea in addition to the progressive ataxia.
- Growth and sexual development can be delayed and mild learning difficulties are common but not invariable.
- The typical skin and eye lesions usually develop between the ages of 3 and 6 years and are best seen on the conjunctival surface and the ear.
- AT is associated with abnormalities of both humoral and cell-mediated immunity and is caused by mutations in the *ATM* gene.
- Clinically related conditions xeroderma pigmentosum and Cockayne's syndrome are also caused by defects in DNA repair; they are much rarer and associated with additional features, most frequently skin abnormalities.

Conjunctival telangiectasia in a patient with ataxi telangiectasia



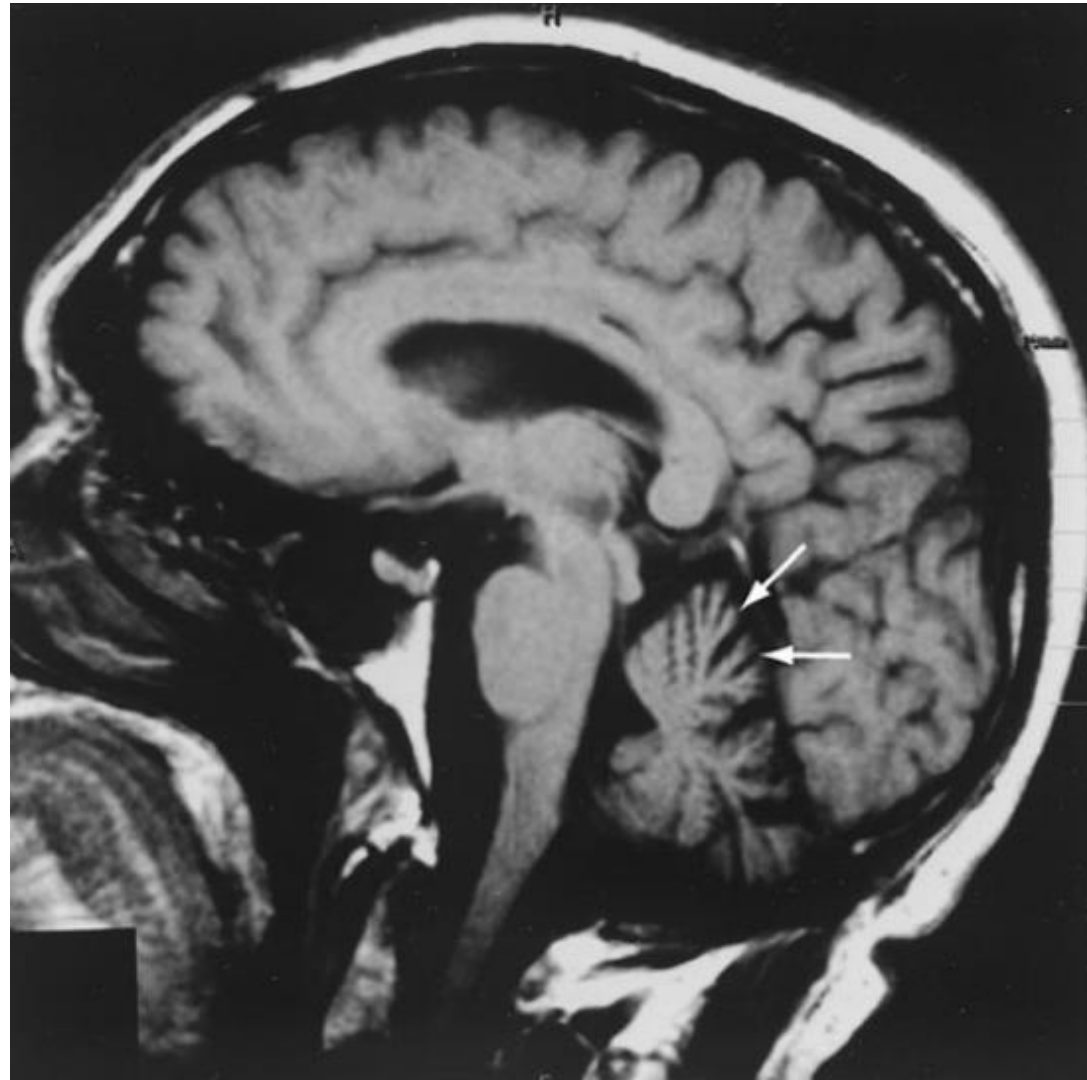
Autosomal dominant cerebellar ataxias

- This is clinically and genetically a heterogeneous group of disorders.
- There are currently some 30 identified loci for the dominant ataxias.
- These are numbered in the sequence of each locus as it is found.
- Spinocerebellar ataxia type 1 (SCA 1) was one of the first neurological diseases to have its gene mapped, reported in 1977.
- Not all the genes have been identified.
- Some common themes at both a clinical and molecular level emerge.
- The molecular consequences of the most common type of mutation is an expanded CAG repeat.

SCA no.	Genetic locus	Gene	Clinical pointers	Notes
1	6p22.3	Ataxin 1 CAG repeat	Complicated, including: neuropathy, cognitive decline, pyramidal and extrapyramidal features	Probably most common worldwide
2	12q24.13	Ataxin 2 CAG repeat	As above, plus slow saccades and neuropathy	
3	14q32.12	Ataxin 3 CAG repeat	As above but a stronger extrapyramidal phenotype	
4	16q24-qter	N/K	Associated with a neuropathy	
5	11q13.2	SPTBN2 beta-III spectrin D		
6	19p13.13	CACNA1A CAG repeat	Later onset (>45 years) and relatively pure	Point mutations in this gene produce either familial hemiplegic migraine or EA2
7	3p14.1	Ataxin 7 CAG repeat	Complicated, with slow saccades and prominent macular dystrophy	Shows the most marked anticipation in the known SCAs
8	13q21	Kelch-like 1 CTG repeat		Uncertain pathogenicity
9	SCA9	Reserved		
10	22q13.31	Ataxin 10 ATTCT repeat		
11	15q14–q21.3	Tau-tubulin kinase 2 point mutation		Pure ataxia
12	5q32	PPP2R2B CAG repeat	Prominent tremor	
13	19q13.33	KCNC3 point mutations		
14	19q13.42	PRKCG point mutations	Prominent dystonia	
15	3p24.2-pter	ITPR 1		

16	8q23–q24.1	Not known (NK)		
17	6q27 TBP	TBP CAG repeat	HD-like features	Rare
18	7q31–q32	N/K		Additional sensor-motor neuropathy
19	1p21–q21	KCND3		Allelic to SCA 22
20	11q12	Heterozygous duplication		A contiguous gene duplication syndrome
21	7p21.3-p15.1	TMEM230	Cognitive Impairment	
22	1p21–q23	KCND3		Allelic to SCA 19
23	20p13-p12.2	PDYN		
24	1p36	N/K		1 Family
25	2p21-p15	N/K		1 Family
26	19p13.3	EEF2		1 Family
27	13q33.1	FGF14 point mutations		
28	18p11.22–q11.2	AFG3L2		Can also cause AR spastic ataxia 5
Unspecified	16q22.1	Puratrophin 1 point mutations		Overlaps SCA 4 region, but not definitively SCA 4 gene

Cerebellar atrophy (*arrows*). MRI, SCA1, spinocerebellar ataxia type 1



Sagittal MRI of the brain and spinal cord of a patient with Friedreich's ataxia, demonstrating spinal cord atrophy

